

Anomaly Detection

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Anomaly detection example

Aircraft engine features:

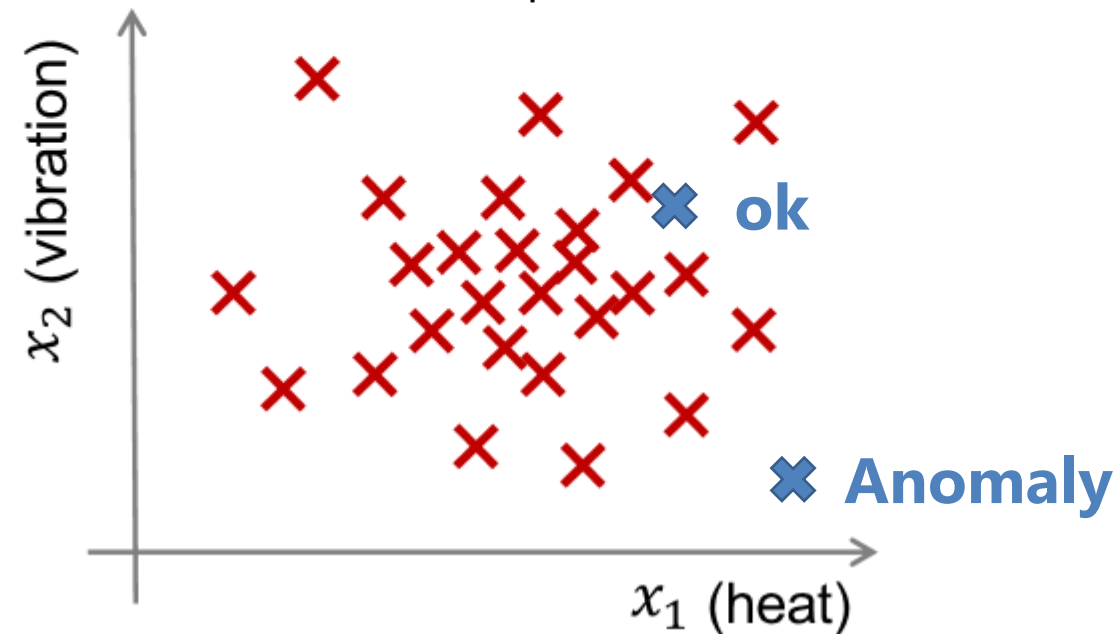
x_1 = heat generated

x_2 = vibration intensity

...

Dataset: $\{x^{(1)}, x^{(2)}, \dots, x^{(m)}\}$

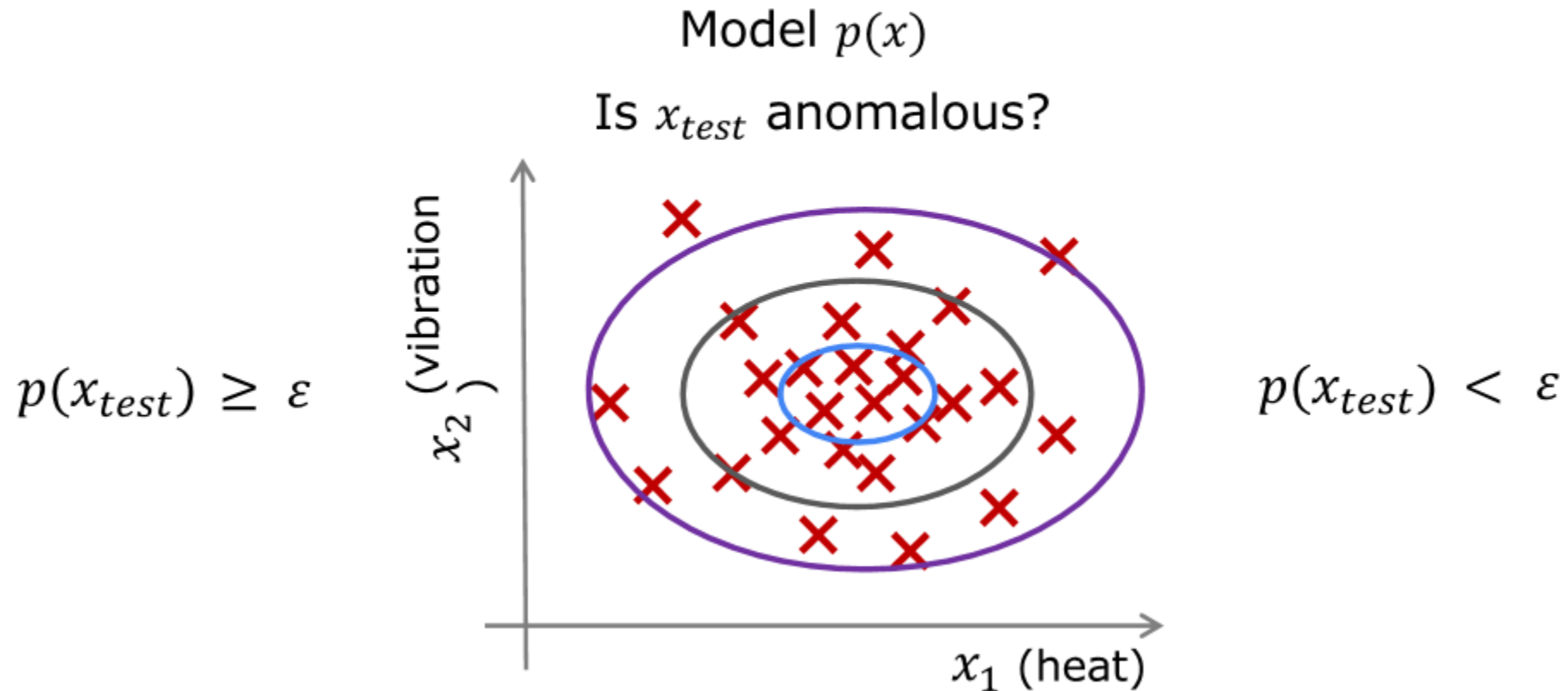
New engine: x_{test}



Density Estimation

Probability of x being seen in the dataset

Dataset: $\{x^{(1)}, x^{(2)}, \dots, x^{(m)}\}$



Anomaly detection example

Fraud detection:

- $x^{(i)}$ = features of user i 's activities
- Model $p(x)$ from data.
- Identify unusual users by checking which have $p(x) < \varepsilon$

- How often log i?
- How many web pages visited?
- N. Transactions?
- N. posts?

- To perform additional checks to identify real fraud vs false alarms

Manufacturing:

$x^{(i)}$ = features of product i

- Airplane engine
- Circuit board
- Smartphone

Monitoring computers in a data center:

$x^{(i)}$ = features of machine i

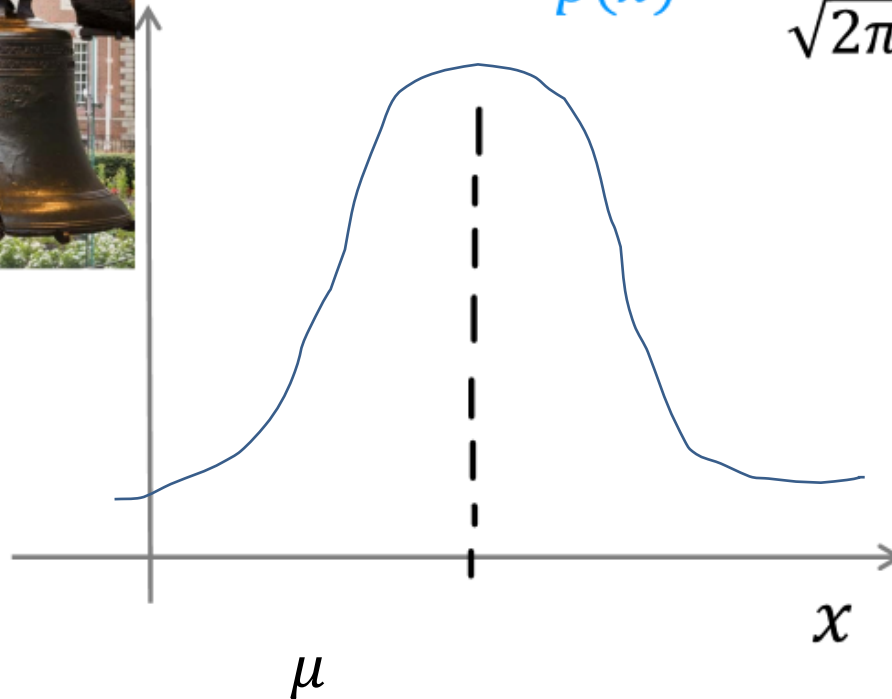
- x_1 = memory use,
- x_2 = number of disk accesses/sec,
- x_3 = CPU load,
- x_4 = CPU load/network traffic.

Gaussian (Normal) Distribution

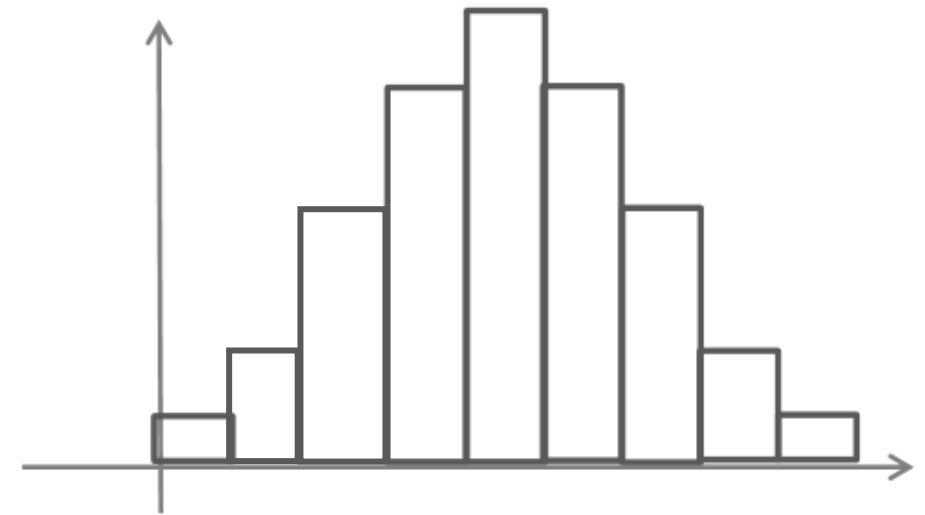
Gaussian (Normal) distribution

Say x is a number.

Probability of x is determined by a Gaussian with mean μ , variance σ^2 .

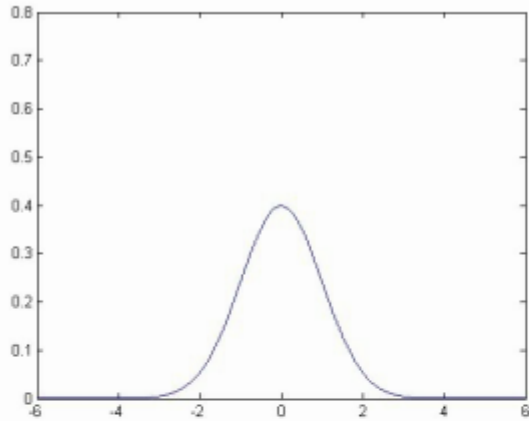


$$p(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{\frac{-(x-\mu)^2}{2\sigma^2}}$$

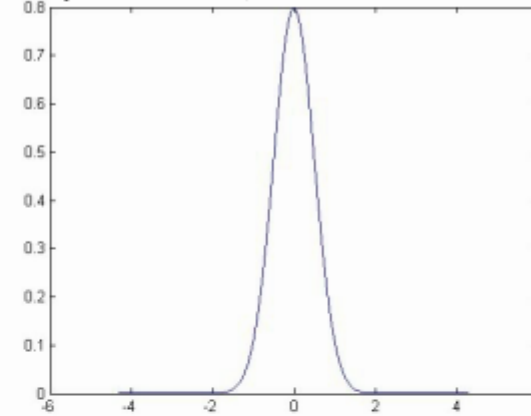


Gaussian distribution example

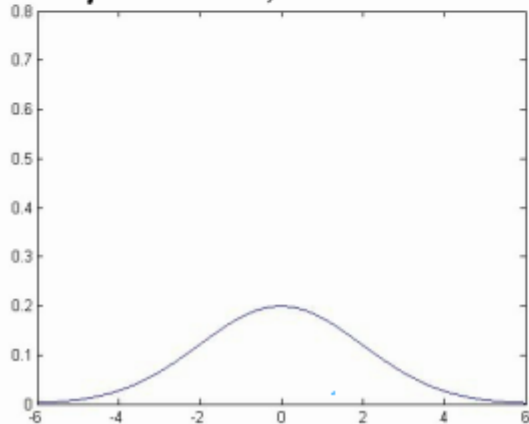
$$\mu = 0, \sigma = 1$$



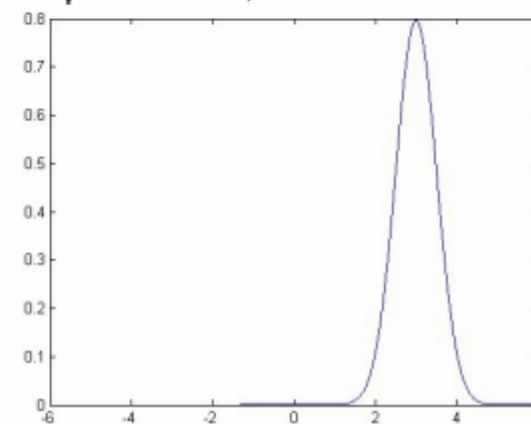
$$\mu = 0, \sigma = 0.5$$



$$\mu = 0, \sigma = 2$$



$$\mu = 3, \sigma = 0.5$$

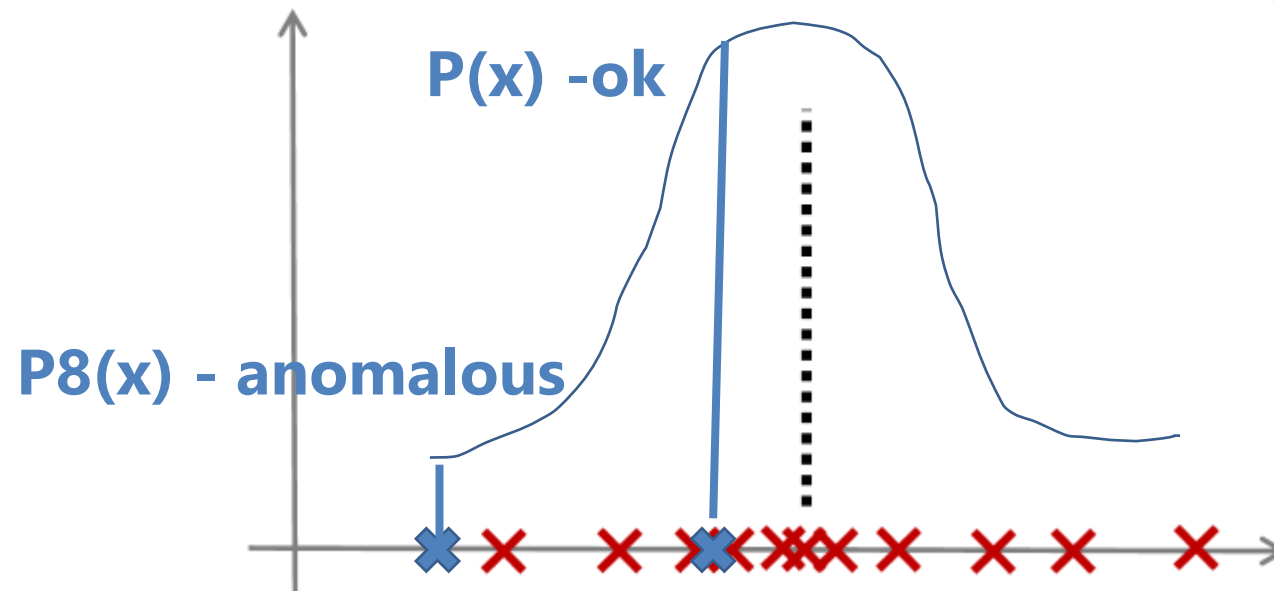


Parameter estimation

Dataset: $\{x^{(1)}, x^{(2)}, \dots, x^{(m)}\}$

$$\mu_j = \frac{1}{m} \sum_{i=1}^m x_j^{(i)}$$

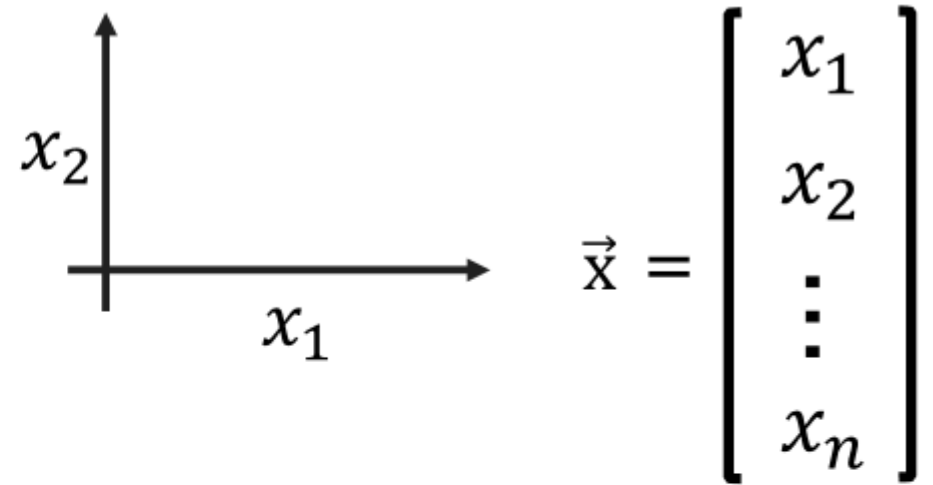
$$\sigma_j^2 = \frac{1}{m} \sum_{i=1}^m (x_j^{(i)} - \mu_j)^2$$



Algorithm

Density estimation

Training set: $\{\vec{x}^{(1)}, \vec{x}^{(2)}, \dots, \vec{x}^{(m)}\}$
Each example $\vec{x}^{(i)}$ has n features


$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

$$p(\vec{x}) = p(x_1; \mu_1, \sigma_1^2) * p(x_2; \mu_2, \sigma_2^2) * p(x_3; \mu_3, \sigma_3^2) * \dots * p(x_n; \mu_n, \sigma_n^2)$$

$$= \prod_{j=1}^n p(x_j; \mu_j, \sigma_j^2)$$

$$p(x_1 = \text{high temp}) = 1/10$$

$$p(x_2 = \text{high vibra}) = 1/20$$

$$p(x_1, x_1) = p(x_1) * p(x_2)$$

Anomaly detection algorithm

1. Choose n features x_i that you think might be indicative of anomalous examples.
2. Fit parameters $\mu_1, \dots, \mu_n, \sigma_1^2, \dots, \sigma_n^2$

$$\mu_j = \frac{1}{m} \sum_{i=1}^m x_j^{(i)} \quad \sigma_j^2 = \frac{1}{m} \sum_{i=1}^m (x_j^{(i)} - \mu_j)^2$$

Vectorized formula

$$\vec{\mu} = \frac{1}{m} \sum_{i=1}^m \vec{x}^{(i)} \quad \vec{\mu} = \begin{bmatrix} \mu_1 \\ \mu_2 \\ \dots \\ \mu_n \end{bmatrix}$$

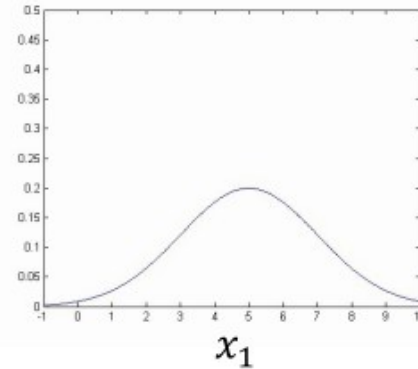
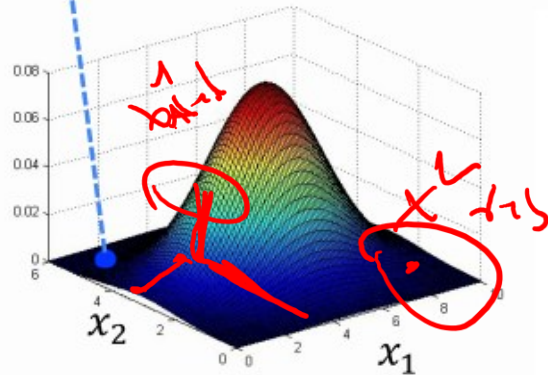
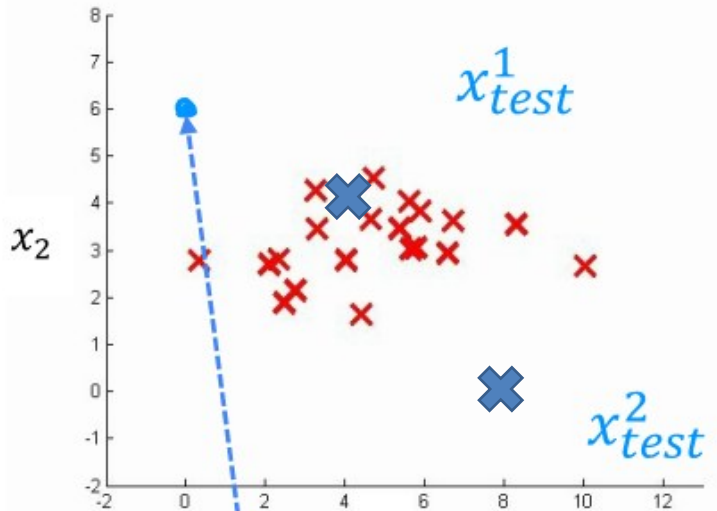
3. Given new example x , compute $p(x)$:

$$p(x) = \prod_{j=1}^n p(x_j; \mu_j, \sigma_j^2) = \prod_{j=1}^n \frac{1}{\sqrt{2\pi}\sigma_j} \exp\left(-\frac{(x_j - \mu_j)^2}{2\sigma_j^2}\right)$$

Anomaly if $p(x) < \varepsilon$

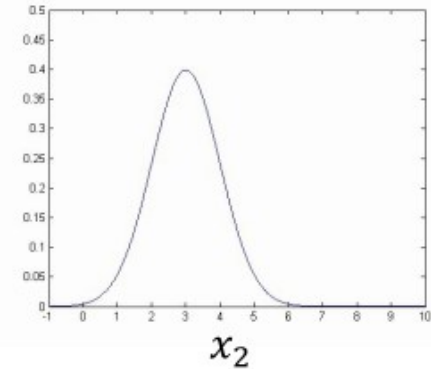
Note: if $p(x_j; \mu_j, \sigma_j^2)$ is small $\rightarrow p(x)$ is small

Anomaly detection example



$$\mu_1 = 5, \sigma_1 = 2$$

$$p(x_1; \mu_1, \sigma_1^2)$$



$$\mu_2 = 3, \sigma_2 = 1$$

$$p(x_2; \mu_2, \sigma_2^2)$$

$$\varepsilon = 0.02$$

$$p(x_{test}^{(1)}) = 0.0426$$

ok

$$p(x_{test}^{(2)}) = 0.0021$$

Anomaly

Anomaly detection vs. supervised learning

Anomaly detection vs. Supervised learning

Very small number (0 to 20) of positive examples $y = 1$;
large number of negative examples ($y = 0$);
Model $p(x)$ with just negative examples.
Use positive examples for cv and test sets

Many different “types” of anomalies.
Hard for any algorithm to learn
(from just positive examples)
what the anomalies look like.
Future anomalies may look
nothing like any of the anomalous
examples seen so far.

Example: Fraud

Large number of positive
and negative examples.

Enough positive examples
for algorithm to get a sense of
what positive examples are like.
Future positive examples likely to be
similar to ones in training set.

Example: Spam

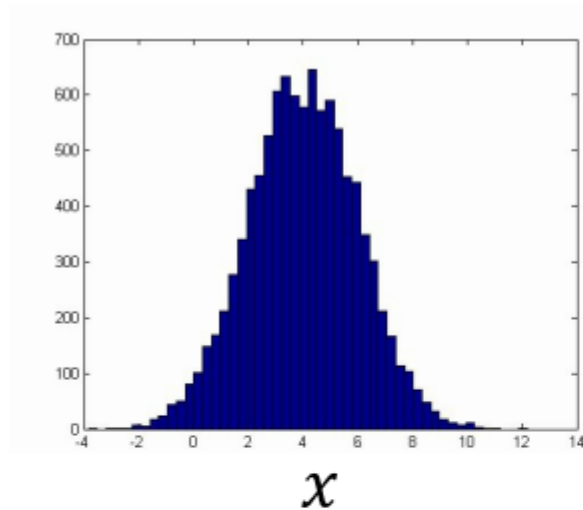
Anomaly detection vs. Supervised learning

- Fraud detection
- Manufacturing:
Finding new previously unseen defects.
(e.g. aircraft engines)
- Monitoring machines in a data center
- Security applications

- Email spam classification
- Manufacturing:
Finding known, previously seen defects
(e.g. scratches on smartphones, $y = 1$)
- Weather prediction (sunny/rainy/etc.)
- Diseases classification

Choosing what features to use

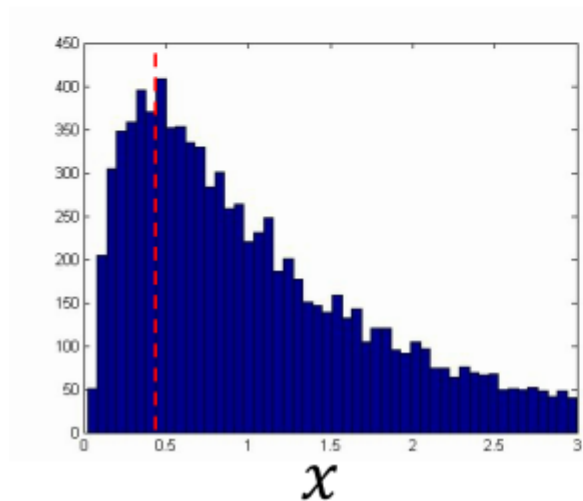
Non-gaussian features



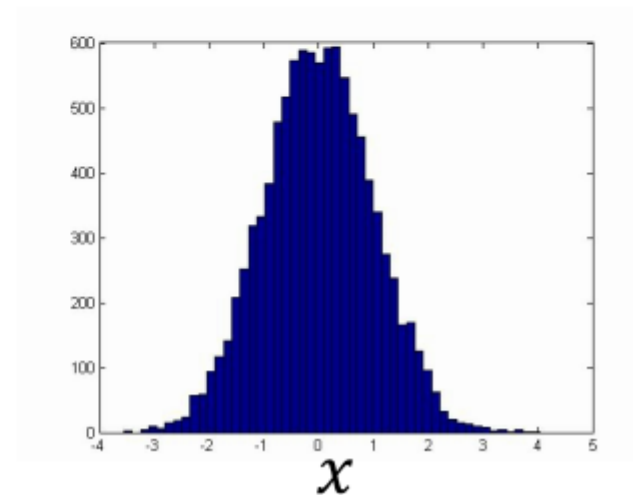
$$p(x_1; \mu_1, \sigma_1^2)$$

plt.hist(x)

$$\begin{array}{ll} x_1 & \log(x_1) \\ x_2 & \log(x_2 + 1) \\ x_3 & \sqrt{x_3} = x_3^{1/2} \\ x_4 & x_4^{1/3} \end{array}$$



np.log(x)



Error analysis for anomaly detection

Want $p(x)$ large for normal examples x .
 $p(x)$ small for anomalous examples x .

Most common problem:

$p(x)$ is comparable for normal and anomalous examples.
($p(x)$ is large for both)

